Introducing Continuous Cost Governance

Extending FinOps from Cost Optimization to Financial Accountability

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Executive Summary

Between 2019 and 2026, cloud financial management matured from an improvised practice into a recognised discipline. The FinOps Foundation's framework gave the field a shared vocabulary, a persona model, and a capability map; the FOCUS specification began to normalise billing data across providers; and the major cloud vendors embedded cost optimisation into their reference architectures. The discipline has succeeded on its own terms.

It has also moved. The *State of FinOps 2026* survey of 1,192 practitioners managing more than $83 billion in annual cloud spend reports that 78% of FinOps practices now sit within the CTO or CIO organisation, while the share reporting to the CFO has fallen to 8% (FinOps Foundation, 2026a). Over the same period, the Foundation revised its mission from advancing the people who manage the value of *cloud* to the value of *technology*, and practitioners reported that scope expansion, governance, organisational alignment and forecasting now collectively outweigh optimisation as a priority. FinOps has shifted **up** into technology strategy and **left** into engineering workflow. It has not, on the evidence, shifted **across** into the finance control environment.

That is the seam this paper examines. A technical recommendation becomes a financial result only when someone is accountable for it, someone authorises it, someone implements it, and someone independently measures what actually changed. Those steps are the ordinary substance of internal control in every other category of corporate expenditure. In consumption-based technology spend they are, in most organisations, not systematically performed, and the industry's headline metric, "potential savings," measures the absence of exactly that discipline.

This paper proposes **Continuous Cost Governance (CCG)**: a control specification that treats each optimisation decision as a governed transaction with an owner, an evidentiary record, an authorisation, and an independently verified outcome. It is offered as an extension to FinOps, not a replacement for it, and not as an industry standard.

The paper also states plainly what CCG is not: it is not novel. Nearly every component described here already exists somewhere: in the FinOps Framework's Policy & Governance capability, in Technology Business Management, in COSO's control activities, in ITIL change enablement, in continuous controls monitoring, and in the thirty-year-old measurement and verification protocols of the energy efficiency sector. Section 3.3 catalogues those overlaps honestly. What appears genuinely additive is narrower than a new discipline: a specification for the object being governed, an evidentiary standard for what may be reported as a financial result together with a protocol that makes that standard executable, and a name for an organisational gap that recent survey data now makes visible.

Continuous Cost Governance in one sentence

> **FinOps determines what should be done with technology spend. Continuous Cost Governance establishes what was actually done, on whose authority, and what it verifiably delivered.**

Key Contributions

This paper contributes three things, and claims no more:

1. **A decision-centric governance model.** It proposes the individual cost decision, rather than the period, the resource, or the service, as the unit of governance, with a defined lifecycle and a retained evidentiary record.
2. **A reportability standard for realised savings.** It proposes a rule separating what may be tracked operationally from what may be reported as a financial result, grounded in established measurement and verification practice. Appendix A transposes that practice to consumption-based spend as a working protocol, so that the standard is executable rather than aspirational.
3. **A control specification that extends rather than competes with FinOps.** It locates that specification inside the FinOps Framework's Policy & Governance capability, and names the organisational seam between engineering-owned telemetry and finance-owned control that current practice leaves unattended.

Continuous Cost Governance is abbreviated **CCG** throughout the remainder of this paper.

Who This Paper Is For

This paper is written for those with fiduciary or operational responsibility for technology expenditure:

- **CFOs, controllers and finance business partners** evaluating whether internal control over variable, consumption-based spending is proportionate to its materiality.
- **FinOps practitioners and cloud architects** whose recommendations are not converting into implemented change at the rate their analysis warrants.
- **CIOs, CTOs and infrastructure executives** balancing delivery velocity against budget predictability.
- **Audit committees, internal audit functions and boards** assessing the maturity of governance over cloud, SaaS and AI expenditure now that it rivals labour as a principal cost line in many organisations.

What This Paper Is Not

- **Not an accounting or legal standard.** CCG does not interpret GAAP or IFRS, does not prescribe capitalisation treatment for cloud refactoring labour, and is not legal advice. It may generate evidence useful to compliance activity; it does not constitute compliance.
- **Not a replacement for FinOps.** The FinOps Framework describes how to build a practice that generates and prioritises optimisation. CCG addresses what happens to an individual optimisation decision after it is generated.
- **Not a claim of novelty.** See Section 3.3 and Section 4.
- **Not a product specification.** CCG originated in observations made during the development of a commercial cost governance tool, which now operates an early, partial implementation of the loop's assignment and verification stages. That origin is disclosed here for transparency; the framework is described in vendor-neutral terms and should be evaluated independently of any implementation, and no results from that tool are cited as evidence in this paper.

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1. The Evolution of Cloud Financial Management

The economic shift underlying this discipline is well documented. Capital expenditure on depreciating assets, governed through procurement cycles and approved in advance, was displaced by operating expenditure incurred continuously by engineers through API calls. The buyer moved from finance to engineering. The commitment moved from a signed purchase order to a running instance. Gartner forecast worldwide end-user spending on public cloud services at $723.4 billion for 2025, growing 21.5% year over year, with cloud infrastructure and platform services alone reaching approximately $301 billion (Gartner, 2024).

1.1 What FinOps Solved

FinOps emerged as a practitioner response to that shift. Its contribution is substantial and should not be understated:

- **A shared language.** The FinOps Framework's principles, personas, domains and capabilities gave finance and engineering a common vocabulary where none existed (FinOps Foundation, 2025a).
- **Data normalisation.** The FinOps Open Cost and Usage Specification (FOCUS) established a provider-neutral schema for billing data, addressing a problem that had previously consumed disproportionate practitioner effort (FinOps Foundation, 2026b).
- **Decentralised accountability.** The principle that "everyone takes ownership for their technology usage" moved cost consciousness into engineering teams rather than concentrating it in a central cost-control function.
- **Rate optimisation at scale.** Commitment-based discount instruments (reserved instances, savings plans, and committed use discounts) became a managed portfolio rather than an ad hoc purchase.
- **Vendor reinforcement.** The AWS Well-Architected Framework elevated "Implement Cloud Financial Management" to a design principle of its Cost Optimization pillar (Amazon Web Services, 2024); Microsoft and Google published comparable guidance.

1.2 Where the Discipline Has Moved Since 2025

Two developments are material to the argument of this paper.

First, **scope expanded**. Framework 2025 introduced Scopes as a core element, formalising the "Cloud+" reality that practitioners were already managing SaaS, licensing, private cloud and data centre spend alongside public cloud (FinOps Foundation, 2025a). By 2026, 98% of respondents reported managing AI spend (up from 63% in 2025 and 31% in 2024), alongside 90% for SaaS, 64% for licensing, 57% for private cloud and 48% for data centre (FinOps Foundation, 2026a).

Second, **the practice moved up the technology organisation rather than into finance**. The 78%/8% reporting split noted above is the clearest available signal. The Foundation's own analysis of Fortune 100 practices describes an emerging "FinOps Enabled Executive": a VP- to EVP-level technology leader, typically with 20 to 25 years of engineering background, positioned as a "COO for the CTO" (Storment, 2025). This is a credible and probably correct trajectory for the discipline. It is also, unambiguously, a trajectory *within the technology function*.

Meanwhile practitioners report diminishing returns from the optimisation work that built the discipline. One characterised the position directly: "We have hit the 'big rocks' of waste and now face a high volume of smaller opportunities that require more effort to capture" (FinOps Foundation, 2026a). Another organisation reported reaching 97% of identified optimisation in its cost optimisation hub, with the remaining 3% deliberately not actioned for business reasons; this is a mature outcome, and one that only means something because the organisation could distinguish *actioned* from *identified*.

No published dataset establishes how many organisations can make that distinction reliably. Practitioner testimony, including the example above, suggests many cannot; this paper treats that as a plausible and widely reported condition rather than a measured one, and Section 7 identifies closing that gap as the framework's most pressing empirical need. That gap, the unmeasured space between what tooling identifies and what an organisation can show it actually did, is the problem this paper addresses.

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2. The Remaining Governance Challenge

2.1 Potential Savings Are Not Financial Outcomes

Cloud cost tooling is optimised to produce recommendations. A rightsizing engine surfaces an oversized instance and attaches a dollar figure. That figure is routinely aggregated across an estate, presented as "potential savings" or "cost avoidance," and reported to leadership as an achievement.

It is not an achievement. It is a hypothesis about a counterfactual. A financial outcome exists when a change is implemented, when the reduction survives comparison against a baseline adjusted for business volume and rate changes, and when the resulting figure is reflected in a budget, a forecast, or the ledger. Every step between the hypothesis and the outcome is a step where the value can be, and frequently is, lost.

The FinOps community is candid about this. The measurement problem was raised repeatedly in the 2026 survey: "Once you fix it, it's gone. How do we give developers credit for shift-left activities?" (FinOps Foundation, 2026a). The Foundation's own guidance on usage optimisation notes that estimating savings requires accounting for the full cost of change (labour hours, service disruption, and transformation complexity), not merely the delta in the bill. Procurement disciplines have long distinguished realised savings from cost avoidance and tracked them separately. Cloud cost management, for the most part, has not adopted that distinction.

2.2 Why Recommendations Fail to Become Outcomes

Four mechanisms are observable, and none of them is a tooling failure.

Evidence asymmetry. A recommendation asserts a saving but rarely proves safety. The engineer receiving it bears the entire downside of a production incident and captures none of the upside of a reduced bill. Absent evidence such as sustained utilisation percentiles, headroom against peak, and dependency and blast-radius analysis, declining the recommendation is the individually rational choice. Recommendation volume without evidence produces fatigue, and fatigue produces silent attrition.

Ownership ambiguity. A resource tag identifies a cost centre. It does not identify a person with the authority and the standing to approve a change to a production system. Where the

routing from finding to decision-maker is unreliable, findings accumulate in a queue that nobody owns.

Absent financial context. Engineers cannot reasonably assess whether a rightsizing action will drop utilisation below a committed-use threshold, alter the allocation of shared costs, or affect a unit-economic metric reported externally. That context sits in finance. In most organisations it is not consulted before the change, only after the variance.

No verification loop. Where implemented savings are never independently measured, there is no feedback signal distinguishing accurate recommendations from inaccurate ones, and no basis on which finance can rely on the number. The estimate is never falsified, so it is never improved.

Supporting evidence is directionally consistent but methodologically imperfect, and should be read with care. Flexera's *State of the Cloud* series reports self-estimated wasted cloud spend of 27% in 2025, rising to 29% in 2026, which marks the first increase in five years and is attributed to AI and new service complexity (Flexera, 2026). These are respondent estimates, not audited measurements. McKinsey research based on a survey of more than 800 executives, drawn primarily from manufacturing, found that 74% of cloud transformations failed to deliver expected savings or business value, with 40% of respondents exceeding cloud budgets (McKinsey & Company, 2022; CFO Dive, 2021). The sample limits generalisability. What these sources establish is that a gap is widely perceived; they do not establish its precise magnitude, and this paper does not assert one.

2.3 The CFO Perspective

For finance leadership the issue is not the absolute quantum of cloud waste. It is that a large, volatile, and rapidly growing expense category is being managed through a control environment materially weaker than the one applied to categories of comparable size.

Three consequences follow. **Forecast reliability degrades** when the inputs to the forecast include unverified savings assumptions. **Internal control coverage becomes uneven**: technology spend increasingly rivals labour as a principal cost line in large enterprises (Storment, 2025), yet the delegation of authority, segregation of duties and evidence retention that finance takes for granted in procurement are inconsistently applied to consumption decisions made by engineers. In the language of the Institute of Internal Auditors' Three Lines Model and Global Internal Audit Standards (Institute of Internal Auditors, 2020, 2024), FinOps operates almost entirely as a first-line operational practice within engineering, without a connected second-line financial controllership checkpoint or a third-line testable assurance trail. And **fiduciary reporting weakens**: a board asking whether a stated efficiency programme delivered has, in many organisations, no answer that survives scrutiny, only a dashboard of opportunities.

CFO reporting relationships in this domain declined to 8% in 2026 (FinOps Foundation, 2026a). Whatever else that number means, it does not describe a discipline moving toward the finance control environment.

2.4 Why AI Changes the Calculus

The argument above would be a modest one if consumption-based spend were stable. It is not, and the direction of change is the strongest practical case for the framework this paper proposes.

Artificial intelligence decomposes technology expenditure into a far larger population of far smaller decisions. Tokens, inference calls, embedding generation, vector database operations, GPU reservations, retrieval pipelines and increasingly autonomous agentic workloads each carry cost, and each is initiated at a level of the organisation well below where budgets are set. Where a traditional infrastructure commitment was a discrete, annual, negotiated event with a purchase order attached, an AI workload generates a continuous stream of consumption decisions with no procurement moment at all. The 2026 survey data captures how fast this arrived: 98% of practices now manage AI spend, against 63% a year earlier and 31% the year before that, and practitioners rank visibility into AI cost, allocation of that cost to business units, and determination of AI value as their three leading difficulties (FinOps Foundation, 2026a).

Three properties compound the governance problem. Pricing models are unfamiliar and vary sharply between providers, so cost cannot be inferred from architecture as reliably as it once could. Attribution is harder, because a single model endpoint may serve many business functions. And the value denominator is contested; as one practitioner put it, “Is your AI providing value? No one can answer that question yet” (FinOps Foundation, 2026a).

The consequence is directional rather than dramatic. As AI moves from experiment to material operating expense, the centre of gravity in technology financial management shifts from *annual budgeting*, which governs a small number of large commitments, toward *continuous financial control*, which must govern a large number of small ones. Periodic review scales poorly against that shape of spend. Per-decision control, applied proportionately, scales better. This paper does not claim that CCG is the only response to that shift, but the shift is what makes the response worth specifying now rather than later.

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3. Introducing Continuous Cost Governance

3.1 Definition

> **Continuous Cost Governance (CCG)** is a proposed control specification for consumption-based technology expenditure, in which each material cost decision is treated as a governed transaction (carrying evidence sufficient to support a decision, a named accountable owner, an authorisation proportionate to its risk and materiality, and an independently verified outcome), and in which only verified outcomes are reported as financial results.

The emphasis is deliberate. CCG does not propose new ways to find savings; that problem is well served. It proposes a specification for what happens to a finding once found, and a standard for what may be reported once acted upon.

The central term carries the weight of the framework and is defined precisely:

> A **governed transaction** is a single cost decision that carries, as a retained record: the evidence on which it was proposed, the named party accountable for it, the financial review it received, the authority under which it was approved, declined or deferred, and the independently verified outcome it produced.

A note on scope. This paper uses public cloud examples because that is where the practice, the data and the tooling are most mature. The control specification is not cloud-specific. It applies to any expenditure that is consumption-based, initiated below the level at which budgets are set, and incurred continuously rather than at a procurement moment, which includes SaaS subscriptions, AI and inference services, data cloud platforms, and consumption-metered licensing. It is deliberately **not** extended in this paper to purchase-order-controlled procurement, accounts payable or ERP-mediated spend. Those categories already carry mature control environments (three-way match, purchase-order approval, and established delegation of authority), and the gap this paper identifies does not exist there. Claiming them would broaden the framework's apparent reach while weakening its actual argument.

3.2 The Unit of Governance

The framework's central proposition is a change in the object being governed.

Existing disciplines govern different units. Budget governance governs the *period*. TBM governs the *service or application*. Cloud policy governance governs the *resource* or the *configuration state*. FinOps allocation governs the *cost*. CCG proposes governing the **decision**: a discrete, dated, evidenced, owned, authorised and verified event with a lifecycle and a retained record.

This is a modest-sounding shift with practical consequences. It makes optimisation auditable at the level at which decisions are actually made, it creates a denominator for measuring conversion (findings raised versus findings resolved), and it produces an evidentiary trail in the form finance and audit functions already understand.

3.3 An Honest Assessment of Prior Art

This paper was written under an explicit instruction not to manufacture novelty. Applying that test rigorously, most of CCG is not new.

CCG element	Existing discipline that already covers it	Assessment
Continuous visibility	FinOps *Understand Usage and Cost* domain; FOCUS;	**Not novel.** Fully addressed by existing practice.

CCG element	Existing discipline that already covers it	Assessment
	native provider cost tooling	
Evidence-based findings	FinOps *Workload* and *Usage Optimization* capabilities; provider recommenders; SRE capacity management	**Largely existing.** The extension is the evidentiary *standard* (evidence sufficient to bear an authorisation decision), not the analysis itself.
Clear ownership	FinOps principle "everyone takes ownership"; *Allocation* capability; Weill & Ross decision rights (2004)	**Not novel.** Decision rights and accountability frameworks are foundational IT governance theory.
Finance review	IT Financial Management; TBM taxonomy; COSO control activities	**Not novel as a function.** Modestly novel in its *insertion point*: per-decision and pre-implementation rather than periodic and post-variance.
Human approval	COSO delegation of authority and segregation of duties; ITIL change enablement; SOX IT general controls	**Not novel, and actively contested.** See Section 6.1.
Verified outcomes	IPMVP measurement and verification (EVO, since 1994; 2022) and ASHRAE Guideline 14 (2023); procurement realised-versus-avoided savings; FinOps Effective Savings Rate	**Method not novel.** Its transposition to cloud is the least-solved and most additive element of the framework. Appendix A is this paper's attempt at that transposition.
Executive accountability	TBM (TBM Council, 2025); FinOps *Quantify Business Value* domain; the FinOps Enabled Executive archetype	**Not novel.** The discipline is already moving in this direction.
The composite control loop	**Continuous controls monitoring**: "technologies to reduce business losses through continuous	**Structurally analogous.** CCG is, in substance, continuous controls monitoring applied to

CCG element	Existing discipline that already covers it	Assessment
	monitoring and reducing the cost of audits through continuous auditing of the controls in financial and other transactional applications" (Gartner, n.d.); continuous auditing (CICA/AICPA, 1999; Vasarhelyi et al., 2004); IIA Three Lines Model (IIA, 2020)	consumption-based technology spend across the first, second and third lines.

Two further points of honesty are required.

The terminology is not new. "Cost governance" and "cloud cost governance" are already in general commercial use, defined by multiple vendors and analysts as continuous, policy-driven oversight of technology spend. CCG's contribution, if it has one, is not the phrase. It is the specification behind it.

The academic ground is well trodden. Continuous auditing has a formal definition dating to 1999 and a substantial peer-reviewed literature, principally associated with Vasarhelyi and colleagues at Rutgers. Anyone presenting a "continuous" governance loop to an audit committee should expect that committee to recognise the pattern immediately.

What appears genuinely additive is therefore narrow, and the paper claims no more than this:

1. **The decision as the governed object**, with a defined lifecycle and retained evidentiary record, which is a construct not specified in the FinOps Framework, in TBM, or in cloud policy governance.
2. **A reportability standard** that excludes estimated, potential and avoided savings from financial reporting while permitting them in operational tracking. The FinOps community identifies this measurement problem as unsolved; CCG proposes a resolution rule rather than a metric.
3. **The naming of an organisational seam** that 2026 survey data makes visible: continuous controls monitoring lives in finance and GRC over ERP transactions; FinOps lives in engineering over telemetry; and the space between them (consumption decisions with financial consequence, made by engineers, outside the finance control environment) has no established owner.

CCG is a recombination. Recombinations can be useful. They should not be marketed as inventions.

3.4 The Five Principles of CCG

Version 0.2 of this framework proposed seven principles. On review, those seven conflated two different kinds of statement: durable normative commitments and sequential process steps. The FinOps Framework separates Principles from Phases from Capabilities for precisely this reason, as do COSO and COBIT. Version 1.0 therefore separates them, and adds one principle that was missing.

Principle 1: Cost decisions are governed as transactions, not as reports. The unit of governance is the individual decision, not the monthly bill or the quarterly review. A finding without a lifecycle is a suggestion.

Principle 2: A finding must carry evidence sufficient to bear the decision it requests. The evidentiary burden sits with the party proposing the change, not the party absorbing its risk. Sufficiency scales with consequence: a sustained-utilisation percentile analysis with headroom against observed peak may suffice for a stateless workload; a tier-one transactional system warrants more.

Principle 3: Every finding has a named accountable owner with authority to act. Ownership is a person or a defined role, derived where possible from identity systems rather than inferred from resource metadata alone. A cost centre is not an owner.

Principle 4: Control is proportionate to materiality and risk. Governance applied uniformly to all decisions is governance that will be circumvented. Most findings should flow through lightweight, automatable paths; formal review and separated authorisation should be reserved for decisions that are financially material, operationally risky, or both. This is not merely a concession to practicality: DORA identifies “treating all changes equally” as a distinct pitfall of change approval processes, on the grounds that undifferentiated review prevents reviewers from devoting attention to the changes that actually warrant it (DORA, n.d.). This principle is the one most likely to determine whether CCG is adopted or resented.

Principle 5: Only verified outcomes are reported as financial results. Estimated, potential and avoided savings may be tracked, and are useful operationally. They are not reported to finance, to executives, or to the board as achieved results. Verification means comparison against a baseline adjusted for business volume, rate changes and seasonality.

Crosswalk from Version 0.2. No prior content is discarded. *Continuous Visibility* becomes a precondition of the loop rather than a principle, because it is a data capability and is already fully specified by FinOps and FOCUS. *Evidence-Based Findings* becomes Principle 2. *Clear Ownership* becomes Principle 3. *Finance Review* and *Human Approval* become stages of the loop, governed normatively by Principles 1 and 4; “Human Approval” is renamed **Authorized Change**, because the salient property is authority, not the presence of a human. *Verified Outcomes* becomes Principle 5. *Executive Accountability* becomes the reporting and reallocation stage of the loop, governed by Principle 5.

3.5 The Continuous Governance Loop

The loop retains the seven-stage structure of Version 0.2, now expressed as process rather than principle, and with the exception paths made explicit, because a control with no documented rejection path is not an auditable control.

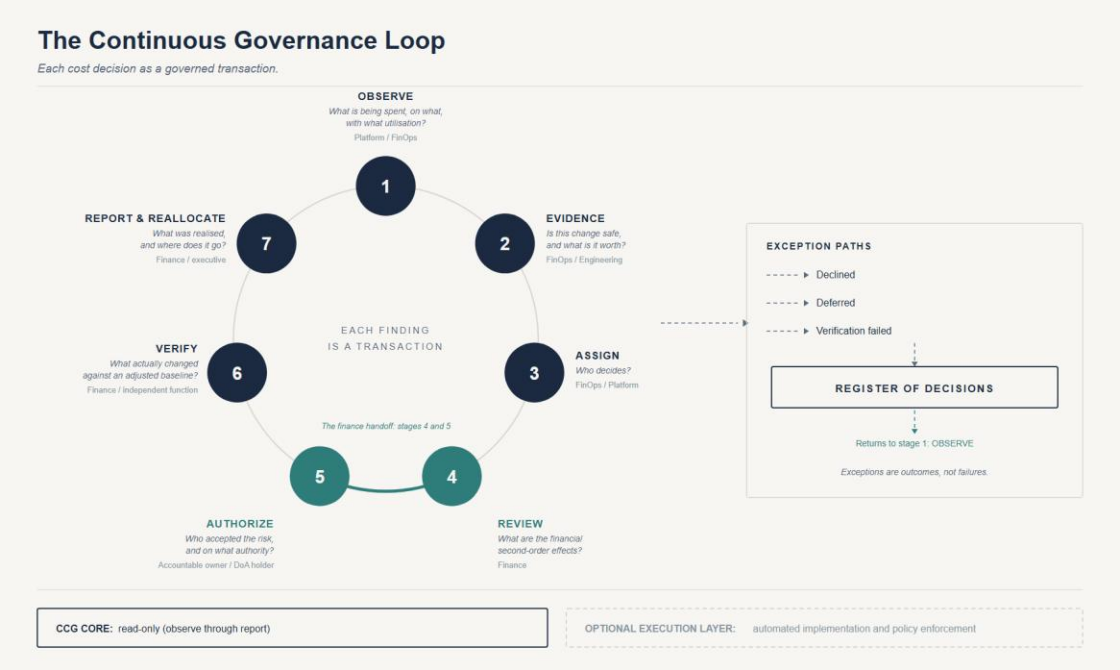


Figure 1: The Continuous Governance Loop. Stages 4 and 5 constitute the finance handoff; Declined, Deferred and Verification Failed are recorded as first-class outcomes in the Register of Decisions.

Stage	Question answered	Primary owner
1. **Observe**	What is being spent, on what, with what utilisation?	Platform / FinOps
2. **Evidence**	Is this change safe, and what is it worth?	FinOps / Engineering
3. **Assign**	Who decides?	FinOps / Platform
4. **Review**	What are the financial second-order effects?	Finance
5. **Authorize**	Who accepted the risk, and on what authority?	Accountable owner / DoA holder
6. **Verify**	What actually changed against an adjusted baseline?	Finance / independent function
7. **Report & Reallocate**	What was realised, and where does it go?	Finance / executive

Three exception paths are first-class outcomes, not failures:

- **Declined:** The owner or finance rejects the change, with a recorded reason. A well-run estate will decline a meaningful share of findings; the 97%/3% example cited earlier is a mature outcome precisely because the 3% was a decision rather than a gap.
- **Deferred:** The change is valid but sequenced behind other work, with a review date.
- **Verification failed:** The change was implemented, but the expected saving did not materialise. This is the most informative outcome in the loop and the one most systems discard. It is the feedback signal that improves estimate quality.

The loop is designed to function without infrastructure write access. **CCG Core** (observe, evidence, assign, review, authorize as a recorded decision, verify, and report) requires only read access to cost and telemetry data plus a system of record for decisions. An **optional execution layer** may automate implementation of low-risk actions or enforce policy in CI/CD, but the governance framework is complete without it. This distinction matters to conservative audit and infrastructure functions and was a deliberate design constraint.

Routing and automation. CCG does not require human review of every finding, and a reading that assumes it does will conclude, correctly, that the framework cannot scale. Principle 4 governs the loop's throughput: findings should be routed automatically according to materiality, risk classification and policy, with the substantial majority resolving through automated or lightweight paths and never reaching a review queue. Stages 3 to 5 collapse into a single automated disposition for a low-value change to a non-production resource; they remain distinct, evidenced and separately authorised for a material change to a tier-one system. What the framework requires is not that a human touches every decision, but that every decision carries a record of how it was disposed of and under what authority, including when that authority was a policy rather than a person. Automation is therefore a precondition of CCG at scale, not a departure from it.

3.6 The Governance Pyramid

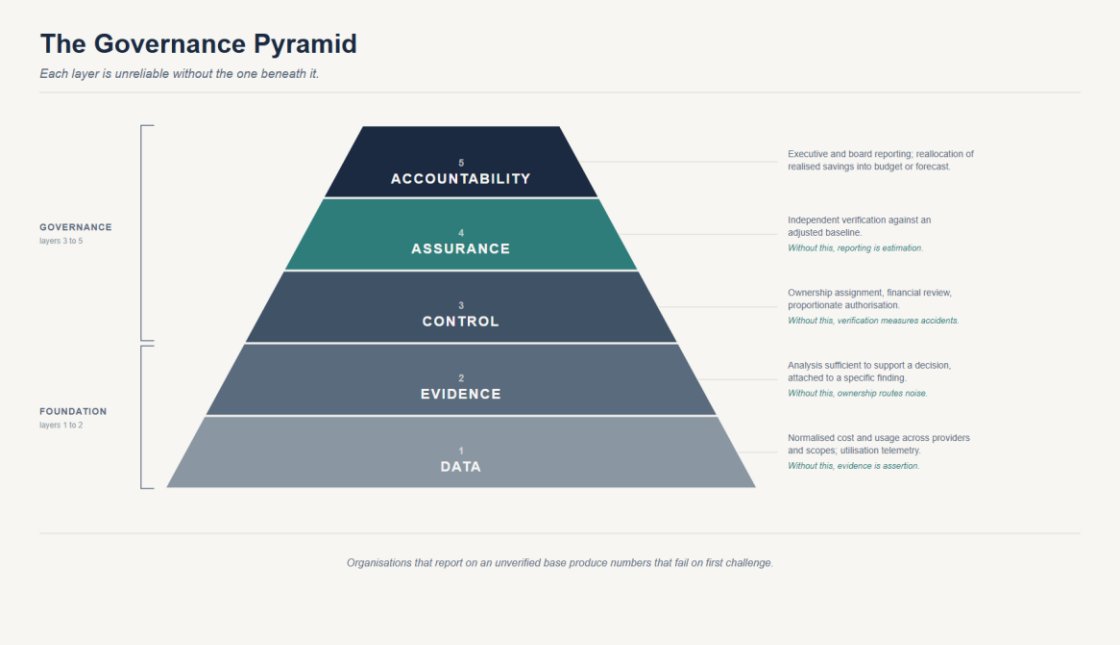


Figure 2: The Governance Pyramid. Each layer depends on the integrity of the layer beneath it.

CCG can be represented as five dependent layers, in which each layer is unreliable without the one beneath it:

1. **Data:** Normalised cost and usage across providers and scopes, together with utilisation telemetry. *Without this, evidence is assertion.*
2. **Evidence:** Analysis sufficient to support a decision, attached to a specific finding. *Without this, ownership routes noise.*
3. **Control:** Ownership assignment, financial review, and proportionate authorisation. *Without this, verification measures accidents.*
4. **Assurance:** Independent verification against an adjusted baseline. *Without this, reporting is estimation.*
5. **Accountability:** Executive and board reporting, and reallocation of realised savings into budget or forecast.

The layered dependency is the argument for sequencing: organisations that attempt executive reporting on an unverified base produce numbers that fail on first challenge, which is the most common way credibility is lost.

3.7 The CCG Maturity Model

The FinOps Foundation's Crawl/Walk/Run model already assesses maturity across its capability set, and its caution applies here too: the objective is the appropriate level of maturity per capability, not uniform advancement (FinOps Foundation, 2025b). The model

below is offered not as a replacement, but as a finer-grained view of two axes (control and verification) that Crawl/Walk/Run treats only lightly.

Level	Visibility	Ownership	Control	Verification	Reporting
1. Reactive	Fragmented billing data	Unassigned	None	None	Ad hoc
2. Visible	Centralised billing	Tag/metadata based	Manual alerts	Estimated avoidance	Periodic dashboards
3. Optimising	Cost plus utilisation	Team-level mapping	Advisory workflow	Simple before/after	Variance narrative
4. Governed	Normalised multi-scope	Identity-derived, named	Risk-proportionate authorisation with recorded decisions	Baseline-adjusted M&V	Reviewable evidence
5. Continuous	Real-time unit economics	Federated with automated routing	Integrated with GRC and change systems	Continuous M&V with estimate calibration	Reconciled to budget and forecast

The transition from Level 3 to Level 4 is where most of the difficulty lies, and it is organisational rather than technical: it requires finance to accept a role in a workflow it does not currently own, and engineering to accept a recorded decision point it does not currently have.

3.8 A Worked Example

> **Illustrative only.** The figures below are synthetic. They are included to demonstrate the method and the arithmetic of verification, not to report a result. No claim is made that these values are typical.

Consider a single finding as it passes through the loop.

1. Observe. Continuous visibility flags a production application tier: fourteen general-purpose instances, thirty-day mean CPU utilisation 11%, mean memory utilisation 24%. The cost tool attaches a naive rightsizing estimate of **$24,100 annualised**.

2. Evidence. Analysis attaches to the finding a ninety-day utilisation distribution (P95 CPU 34%, P99 61%), a dependency map showing two downstream services, and the observation that the P99 peak occurs within a four-hour window on the last business day of each month, correlated with financial close. The proposed target instance size provides roughly twice the headroom against observed P99 for eleven of the fourteen instances. Three instances carry the month-end path and do not clear the threshold.

*Evidence has already changed the proposal: eleven instances, not fourteen. The three excluded are recorded as **Declined**, with the reason attached, rather than disappearing.* Revised estimate: **$18,900**.

3. Assign. Resource metadata resolves to a cost centre, which is not a person. An identity lookup resolves the owning team and its engineering manager as the accountable owner. The finding is routed to a named individual with authority to act.

4. Review. Finance identifies a second-order effect invisible to the engineering analysis: the account is substantially covered by a three-year compute commitment, and withdrawing on-demand consumption from this instance family would push utilisation of that commitment below its break-even point, transferring cost to an unused reservation rather than removing it from the P&L. Finance does not decline the change. It directs that the change be either sequenced after the commitment expires, or offset by migrating a currently uncommitted workload into the freed capacity. The team selects the offset.

This is the stage most commonly absent in practice, and in this example it is the stage that prevents a reported saving from being an accounting illusion.

5. Authorize. The change exceeds the organisation's automatic-routing threshold on two counts (annualised value and tier-one service classification), so it does not resolve through the automated path. The accountable owner authorises it. The record captures who authorised, when, on what evidence, and what risk was accepted.

6. Verify. Ninety days after implementation, the outcome is measured against a baseline adjusted for three effects in the period that are not the authorised change: a 6% decline in transaction volume, a provider list-rate reduction on the in-scope instance family, and the commitment reallocation directed at stage 4. Appendix A sets out the protocol, the equation and the full arithmetic behind the figures below.

	Annualised
Naive estimate at finding	$24,100
Estimate after evidence (11 of 14 instances)	$18,900
Gross observed reduction	$19,400
Less: demand normalisation (transaction volume −6%, measured elasticity)	($450)
Less: provider list-rate reduction	($1,850)
Less: commitment reallocation effect (non-routine)	($1,150)
Verified realised saving	**$15,950**

7. Report & Reallocate. $15,950 is reported to finance and carried into the next forecast cycle. **$24,100 is not reported at any point.** The gap between the two figures, roughly a

third of the original estimate, is not error or waste; it is the difference between a hypothesis and a measured outcome, and under Principle 5 only the latter is a financial result.

What the example is intended to show. Four of the seven stages changed the answer. Evidence removed three instances from scope. Finance review prevented a change that would have transferred cost rather than eliminated it. Baseline adjustment removed $3,450 of apparent saving attributable to factors other than the change itself. And the decision now exists as a record that an auditor, a controller or a successor can reconstruct. The framework's value is not that it found the saving, as the cost tool did that in stage 1, but that it established which part of the saving was real.

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4. How CCG Extends FinOps and TBM

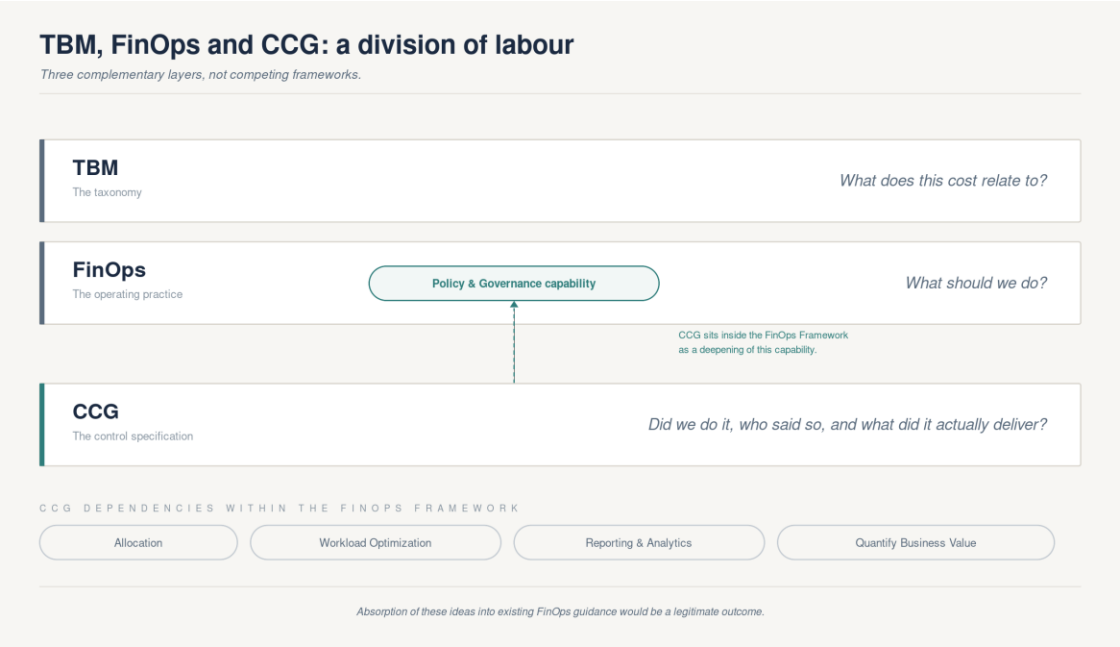


Figure 3: TBM, FinOps and CCG: a division of labour. CCG sits within the FinOps Framework as a deepening of the Policy & Governance capability.

A simple division of labour describes the relationship without diminishing any of the three.

TBM provides the taxonomy: the mapping of technology cost to services, applications, value streams and business capabilities, recently expanded in TBM Taxonomy 5.0 to incorporate dedicated cloud service cost pools and AI solutions (TBM Council, 2025). It answers *what does this cost relate to?*

FinOps provides the operating practice: the culture, capabilities and analysis that generate and prioritise optimisation across an expanding set of scopes. It answers *what should we do?*

CCG proposes the control specification: the evidentiary, authorisation and verification requirements applied to each decision. It answers *did we do it, who said so, and what did it actually deliver?*

Within the FinOps Framework, CCG sits most naturally as a deepening of the **Policy & Governance** capability (renamed from "Cloud Policy & Governance" in Framework 2025; FinOps Foundation, 2025a), with dependencies on Allocation, Workload Optimization, Reporting & Analytics, and Quantify Business Value. If the FinOps Foundation were to absorb these ideas into that capability's guidance rather than as a separate framework, that would be a legitimate and arguably preferable outcome. This paper does not require CCG to exist as a distinct discipline in order for its argument to be useful.

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5. Implications

5.1 For Finance Leaders

The practical ask is narrow: accept a defined, bounded role at stage 4 of the loop for decisions above a materiality threshold that finance itself sets. In return, finance gains a defensible realised-savings figure, an evidentiary trail in a familiar form, and an input to forecasting that has been tested rather than assumed. The threshold is the critical design choice. Set too low, finance becomes a bottleneck on trivial decisions and the process collapses. Set sensibly, the reviewed population should be small by count and large by value, following the ordinary distributional logic of any materiality threshold. This paper does not propose specific figures; no published dataset supports one, and organisations should derive their own from their own finding distribution.

Who staffs stage 4. This is the most common practical question the framework raises, and it does not require a new headcount line in most organisations. Stage 4 is a review, not a build: it asks whether a proposed change interacts with a commitment, an allocation rule, or an externally reported metric, which are questions a FinOps analyst, a technology finance business partner, or an existing FP&A function already fields informally today. What CCG adds is not a role but a checkpoint: the review happens before implementation rather than after the variance, and it is recorded rather than remembered. Organisations already running a FinOps practice with a finance-facing capability can typically staff stage 4 from within it; the change is procedural, not organisational. Only where the reviewed population is large enough to require dedicated capacity (itself a function of the materiality threshold discussed above) does this become a staffing decision rather than a reallocation of existing review effort.

5.2 For Engineering

The value proposition is inverted from how governance is usually presented. CCG's Principle 2 places the evidentiary burden on the party proposing the change, not the engineer absorbing its risk. Correctly implemented, it should *reduce* the volume of

unevidenced recommendations reaching engineering queues and provide documented cover for changes that are made. Incorrectly implemented as a universal approval gate, it will reproduce exactly the failure mode Section 6.1 describes, and engineering's objection will be correct.

5.3 For CIOs and CTOs

The 78%/8% reporting split in Section 1.2 is not only a measurement fact; it is also the shape of a political question that this paper should not pretend away. FinOps already reports into the technology organisation in most enterprises. CCG's stage 4 asks the CIO to route a defined class of decisions through a finance checkpoint before implementation, which can read, at first pass, as ceding a control point that currently sits inside technology.

The more accurate reading is the reverse. An unverified savings number is a liability in a budget conversation, not an asset: it can be, and often is, discounted or disputed by finance precisely because no one outside the technology organisation vouches for it. A verified figure, produced through a review the CIO's own organisation retains operational ownership of, is a number finance has already agreed to. That is leverage in the next budget cycle, not a loss of authority in this one. What CCG asks the CIO to give up is the unilateral right to assert a savings figure without review; what it offers in exchange is a figure finance cannot later unwind. For a function whose credibility is periodically tested by exactly that kind of dispute, the trade is favourable.

5.4 For Boards and Audit Committees

The relevant question is not "how much did we save?" but "on what basis is that figure stated?" A board-appropriate view of technology cost governance would show: findings raised and their disposition; realised savings measured against an adjusted baseline, stated separately from estimates; the population of decisions subject to formal authorisation and the exceptions to it; and the reallocation of realised savings into budget or forecast. Three or four figures, each of which can survive a follow-up question. Estimated savings, aggregate potential, and cost avoidance do not belong in that view.

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6. Limitations and Counterarguments

Intellectual honesty requires that the strongest objections be stated rather than answered pre-emptively.

6.1 The change-approval objection is evidence-based and serious

DORA's research programme has consistently found that formal change management requiring approval from an external body, such as a change advisory board or senior manager, is associated with *worse* software delivery performance. Respondents whose organisations used such processes were 2.6 times more likely to be low performers, and DORA found no evidence that a more formal external review process was associated with

lower change failure rates (DORA, 2019; DORA, n.d.). DevOps practice spent fifteen years dismantling approval gates on the strength of that finding.

This paper does not dispute it. Principle 4 exists because of it. But the tension is not fully resolved, and a candid framework should say so. CCG's position is that DORA's finding concerns *external approval of technical change*, while CCG's stage 5 concerns *authorisation of financial consequence*, and that where the two coincide, peer review captured in the development platform is an acceptable authorisation mechanism, as DORA itself suggests for segregation of duties. Whether that distinction survives contact with practice is an empirical question this paper cannot settle.

6.2 The materiality objection

For many organisations, cloud spend is not material to the consolidated P&L, and the governance overhead CCG proposes would not be proportionate. CCG is likely to be relevant only where technology spend is large in absolute terms, volatile, or subject to specific regulatory or margin-reporting pressure. It should not be presented as universally applicable.

6.3 The verification problem is not solved

There is no established equivalent of IPMVP for cloud. Baseline adjustment for business volume, rate changes and seasonality is conceptually straightforward and practically difficult: cloud environments change continuously, attribution is contested, and the counterfactual is unobservable. Appendix A proposes a first protocol (an adjusted-baseline equation, four measurement options transposed from IPMVP, normalisation drivers by workload class, attribution rules for overlapping changes, and a worked calculation), and that proposal narrows the gap rather than closing it. It is untested in any organisation, it carries no empirical error bounds, there is no evidence that two analysts applying it would agree, and it rules a large class of genuine engineering value out of financial reporting altogether. Section 3.4's Principle 5 therefore still sets a standard that current methods can only partially meet. This remains the framework's largest open weakness and is stated as such.

6.4 Framework proliferation

CIOs face a legitimate objection that the field does not need another framework, another maturity model and another assessment. The response offered here is that CCG is deliberately small (five principles, one loop, and one reportability rule), and that its most likely useful destination is absorption into an existing framework rather than independent survival.

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7. Future Research

Five questions would materially strengthen or falsify the propositions in this paper.

1. **Conversion rate measurement.** What proportion of generated recommendations are implemented, declined, deferred or silently expired, and how does that vary by evidence quality, ownership clarity and organisation size? No credible public dataset currently answers this. It is the empirical foundation the framework lacks.
2. **A measurement and verification protocol for cloud.** Can IPMVP's option structure and adjusted-baseline methodology be adapted to consumption-based technology spend with acceptable error bounds? Appendix A offers a candidate protocol to be attacked rather than adopted, which makes the remaining questions narrower and directly testable: do two independent analysts applying it to the same finding arrive at the same verified figure; what baseline window length is required before cost-elasticity estimates stabilise, by workload class; and what error does the protocol achieve against a known ground truth, such as a deliberately instrumented change with a withheld control group?
3. **The friction/realisation trade-off.** At what materiality thresholds does added authorisation control begin to reduce total realised savings by suppressing implementation? This is directly testable and would resolve the tension in Section 6.1.
4. **Audit utility.** Do internal and external auditors regard per-decision cost governance evidence as useful, and under what circumstances would it affect testing approach or reliance?
5. **AI workload economics.** With 98% of practices now managing AI spend (FinOps Foundation, 2026a) and token- and inference-based pricing lacking established optimisation playbooks, does the CCG loop hold, or do decision cycles compress below the point at which per-decision authorisation is feasible? Section 2.4 argues that AI economics strengthen the case for continuous control; the opposite conclusion, that AI decision velocity defeats per-decision governance entirely, is equally testable and would falsify a central premise of this paper.

This paper is deliberately conceptual: it addresses *why* the control specification is needed and *what* it must contain. A separate companion document addressing *how* (including data structures for the decision record, evidence sufficiency thresholds by workload class, baseline adjustment methods, and reference integrations with change and GRC systems) would be the natural next artefact, and is better written after the questions above have been tested than before.

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8. Conclusion

FinOps solved a problem that genuinely needed solving, and continues to solve harder versions of it. Its trajectory, moving up into technology strategy, left into engineering workflow, and out across an expanding set of technology scopes, is the right trajectory for what it is.

But that trajectory leaves a specific seam unattended. Decisions with financial consequence are made continuously, by engineers, on the basis of estimates, outside the control environment that finance applies to expenditure of comparable size. Continuous Cost Governance proposes a specification for closing that seam: govern the decision, not the report; require evidence proportionate to consequence; name an owner; authorise in proportion to risk; verify against an adjusted baseline; and report only what was verified.

Little of this is new. The components are drawn from internal control, continuous auditing, change management and energy-sector measurement and verification, and the paper has tried to say so plainly. What may be new is the assembly, the object being governed, and the discipline of refusing to report an estimate as a result.

This paper proposes the framework. It does not validate it. Validation requires the empirical work described in Section 7, and the framework should be treated as a hypothesis until that work exists.

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Appendix A: A Measurement and Verification Protocol for Consumption-Based Spend

> **Status.** This appendix is a first attempt at the protocol Section 7 identifies as research question 2. It is offered to be attacked rather than adopted. It has not been applied in any organisation, it carries no empirical error bounds, and it is not a standard. The figures continue the illustrative, synthetic finding from Section 3.8.

A.1 Why the paper requires this

Principle 5 states that only verified outcomes are reported as financial results, and defines verification as comparison against a baseline adjusted for business volume, rate changes and seasonality. Section 6.3 then concedes that no cloud equivalent of IPMVP exists. Those two statements together leave the framework's most distinctive principle without a method, and a principle without a method is a slogan.

The energy efficiency sector resolved the same problem structurally. IPMVP does not measure savings directly, because savings are the difference between actual consumption and what consumption *would have been*, which is unobservable. Instead it specifies a baseline, an adjustment procedure, and a set of measurement options whose rigour is matched to the materiality of the measure (Efficiency Valuation Organization, 2018). That structure transposes. What does not transpose is the content: cloud has no stable physical plant, its configuration changes continuously, its rates change unilaterally, and its commitment instruments couple unrelated workloads together. This appendix specifies the minimum method required to execute Principle 5, and is explicit about where the transposition fails.

A.2 Definitions and the savings equation

Governed scope. The set of resources to which the authorised change applies, rather than the account, the service, or the resources considered and declined. In the Section 3.8 example the governed scope is the eleven instances authorised at stage 5, and excludes the three recorded as Declined.

Measurement basis. Amortised cost, FOCUS-normalised via the normative `EffectiveCost` column (FinOps Foundation, 2026b), rather than invoiced cash (`BilledCost`). `BilledCost` moves when commitments are purchased upfront, which would register a commitment purchase as a cost increase and its amortisation as a saving; unit rate changes (ρ) are similarly evaluated against FOCUS `ListCost` or `ContractedCost`, while

commitment portfolio state is tracked via the FOCUS 1.3 Contract Commitment dataset. Any protocol operating on billed cash will produce artefacts.

Periods. A baseline period preceding the change and a reporting period following it. Both must span at least one complete cycle of every driver that materially moves the governed scope's cost; in the worked example, this requires at least one financial close, because the fleet's peak is a month-end phenomenon. Both are annualised by the same stated method.

The equation.

$$S = B \cdot (D_r / D_b)^e \cdot \rho - R - \Sigma A_{nr}$$

Term	Meaning
S	Verified realised saving, annualised
B	Baseline-period amortised cost (`EffectiveCost`) of the governed scope, annualised
R	Reporting-period amortised cost (`EffectiveCost`) of the governed scope, annualised
D_b, D_r	Demand driver in the baseline and reporting periods
e	Cost elasticity of the governed scope to that driver, **estimated from the baseline window, not assumed**
ρ	Rate factor: reporting-period unit rates divided by baseline unit rates, for the *baseline* configuration
A_nr	Non-routine adjustments: discrete events in the period that are not the authorised change

The terms for demand and rate are the **routine adjustments**: predictable, recurring, modelled. Non-routine adjustments capture migrations, re-architecture, tenancy or account restructuring, commitment portfolio changes, model substitution, and one-off provider credits. The distinction is IPMVP's and it carries the protocol's integrity, because non-routine adjustments cannot be modelled from the baseline and must therefore be enumerated by someone, which is why they are the easiest place to flatter a result.

A.2.1 The elasticity term is where most error enters. Setting the elasticity to 1 (assuming cost moves proportionally with demand) is the default in informal before-and-after comparison, and it is wrong in the direction that matters. Statically provisioned capacity has an elasticity near zero: its cost is determined by a provisioning decision, not by

demand. In IPMVP terms, the multiplicative factor $(D_r/D_b)^e$ is the local constant-elasticity equivalent of a two-parameter linear baseline model $C(D) = C_{\text{fixed}} + c_{\text{var}} \cdot D$, where $e = (c_{\text{var}} \cdot D_b)/B$ is the variable-cost share at baseline (by first-order Taylor expansion, $B \cdot (D_r/D_b)^e \approx B \cdot [1 + e(D_r - D_b)/D_b] = C_{\text{fixed}} + c_{\text{var}} \cdot D_r$). Practitioners may equivalently use the linear fixed-plus-variable specification where demand crosses zero or shifts sharply; what matters is measuring the variable share rather than assuming proportionality. Applying a unit-elasticity normalisation to a static fleet removes genuine saving and reports a smaller number than was achieved. Section A.7.1 quantifies this on the worked example: the error is 12% of the reported figure, from a single unexamined assumption.

A.2.2 Pre-registration of the adjustment set. The adjustment set (which drivers, which elasticity estimate, which non-routine events, and which windows) must be specified and recorded **before** implementation, as part of the stage 4 review. Adjustments selected after the result is known are not verification; they are negotiation. This is the protocol's most important control and among the cheapest to implement, because stage 4 already exists and already produces a record.

A.2.3 Rounding. The worked example rounds line items to the nearest $50 and states so. A protocol that does not fix its rounding convention will produce figures that fail to reconcile across reports, which is a disproportionately common way for an otherwise sound number to lose credibility.

A.3 Four measurement options

Transposed from IPMVP's Options A to D, with rigour matched to materiality per Principle 4.

Option	What is measured	What is stipulated	Cloud application	Reportable under Principle 5?
A: isolation, key parameter	The changed parameter only: resource count or size times amortised unit rate	That the demand envelope is unchanged, within a stated tolerance	High-volume, low-materiality configuration changes resolving on automated paths	Yes, within the tier limits in A.6
B: isolation, all parameters	Full amortised cost of the governed scope in both periods, at resource granularity	Nothing material	A single identifiable finding, which is the default for anything reaching stages 4 and 5	Yes
C: whole-scope	Total amortised cost of an	Resource-level	Diffuse or policy-driven	Only if the detectability

Option	What is measured	What is stipulated	Cloud application	Reportable under Principle 5?
regression	account, subscription or product scope, regressed on drivers	attribution	change across many resources, where isolation is impossible	test in A.7.2 is met
D: calibrated counterfactual	Nothing; the baseline is modelled	The entire baseline	Net-new workloads, prevented spend, and architectural avoidance, where no measured baseline exists	**No.** Operational tracking only

The Option D exclusion is the protocol's sharpest consequence, and it is the mechanism by which Principle 5 excludes cost avoidance. Avoided cost is, by construction, an Option D measurement: it compares actual spend against a counterfactual that was never instrumented. This is not a judgement that avoidance is worthless; preventing a badly provisioned workload from reaching production is often worth more than rightsizing it afterwards. It is a judgement that avoidance cannot be *verified*, and Principle 5 governs reporting, not value. Section A.9 records the cost of that position.

A.4 Normalisation drivers by workload class

The driver must be selected from the cost structure of the workload, not from whichever business metric is nearest to hand. The elasticity column states *a priori* expectations only, offered as hypotheses to be tested per estate; consistent with this paper's practice elsewhere, no measured values are asserted.

Workload class	Primary driver candidates	Expected elasticity (to be measured, not assumed)	Notes
Statically provisioned compute	Transactions, requests, active users	Low	Cost is set by provisioning, not demand. Demand normalisation is small and must be measured.
Autoscaled and	Invocations,	High	The case where unit

Workload class	Primary driver candidates	Expected elasticity (to be measured, not assumed)	Notes
serverless compute	concurrency-seconds, requests		elasticity is defensible, and close to the only one.
Object and block storage	GB-months retained, objects written, retrieval volume	Low for retained, high for retrieval	Retained volume is strongly autocorrelated; regress on level, not change.
Data platform and warehouse	Queries, bytes scanned, slot- or credit-hours	Moderate to high	Query mix shift is a confound; segment by workload before regressing.
AI inference	Input and output tokens, calls, context length distribution	High, but non-linear in context length	Model substitution is a **non-routine** event, not a routine adjustment.
SaaS and metered licensing	Active seats, metered units	Step-wise, not continuous	Tier boundaries break the regression; model as steps, not a slope.

A.5 Attribution when changes overlap

Overlapping changes are the problem Section 7 names, and the one most likely to defeat the protocol in a real estate. Five rules, in precedence order:

1. **Sequence where possible.** Verification windows for findings against the same resources should not overlap. Sequencing is cheaper than attribution.
2. **Isolate where separable.** Where overlapping changes touch disjoint resources, Option B isolation resolves the attribution without further machinery.
3. **Aggregate where inseparable.** Where overlapping changes touch the same resources and cannot be separated, the findings are merged into a single **verification unit** and verified jointly. The joint figure is reported; it is **not** allocated to the constituent findings for financial reporting. Pro-rata allocation by estimate is permitted for operational tracking and estimate calibration only.
4. **Rate and commitment effects belong to the portfolio.** Savings arising from commitment purchases, rate negotiations or discount instrument changes are attributed to the commitment portfolio owner, never to a usage-side finding that happens to coincide with them. Without this rule the same dollar is credited twice

(once to the engineer who reduced usage and once to the analyst who bought the commitment), and the sum of verified savings exceeds the reduction in the bill.

5. **Uniqueness.** The decision register enforces a constraint on resource and period: no resource-period may be credited to more than one verified finding. Where rule 3 merges findings, the merged unit holds the claim.

A.6 Rigour proportionate to materiality

Tier	Trigger	Required option	Minimum windows	Verified by
1	Below the automated-routing threshold	A	30-day reporting against existing telemetry	Automated, with periodic sampled audit
2	Above the routing threshold, non-tier-one service	B	90 / 90	FinOps, with finance sign-off on the pre-registered adjustment set
3	Financially material or tier-one service	B	90 / 90, extended to span a full business cycle	Finance or an independent function
4	Diffuse or programme-level	C	180 baseline / 90 reporting	Finance, with the detectability test documented

Tier 1 is deliberately permissive. A protocol that demands Option B of every finding will be applied to none of them, which is the failure mode Principle 4 exists to prevent.

A.7 Worked calculation

Continuing the Section 3.8 finding. Tier 3, Option B. **All figures synthetic.**

Input	
Governed scope	11 instances authorised at stage 5 (3 Declined, out of scope)
Measurement basis	Amortised cost, FOCUS-normalised
Baseline period	90 days pre-change, spanning two financial closes
Reporting period	90 days post-implementation, spanning two financial closes

Input	
B: baseline annualised	**$38,400**
R: reporting annualised	**$19,000**
Driver	Settled transactions
D_b → D_r	42.0m → 39.5m (**−6.0%**)
e: measured elasticity	**0.19** (standard error 0.07)
ρ: rate factor	**0.951** (a 4.9% list-rate reduction on the in-scope instance family, pro-rated over the window)
A_nr	**$1,150** commitment reallocation (see below)

The elasticity estimate. The elasticity is estimated by ordinary least squares of daily amortised in-scope cost (`EffectiveCost`, which includes fixed instance-hour compute alongside usage-metered network egress and storage I/O attached to the eleven instances) on daily settled transactions across the baseline window, in logs. The regression returns 0.19 with an R^2 of 0.21, explaining only a fifth of daily cost variation. That weak fit is not a defect in the estimate; it *is* the finding. A statically provisioned fleet's cost is largely insensitive to demand (with only its attached transfer and I/O components varying with transaction volume), which is precisely why it was able to sit at 11% mean CPU in the first place.

The non-routine adjustment. Stage 4 directed that freed capacity be backfilled by migrating an uncommitted workload into the three-year commitment. That backfill is a real enterprise benefit, but $1,150 of the governed scope's observed reduction is a *transfer*: committed units the in-scope workload no longer consumes are now consumed by the backfilled workload. Under rule 4 of Section A.5 that benefit is attributed to the commitment portfolio, not to this finding. Crediting it here would double-count it against the backfilled workload's own record.

The calculation.

```
Demand factor          (39.5 / 42.0) ^ 0.19   =  0.98841
Demand-adjusted baseline   38,400 × 0.98841   = 37,954.9
Normalised baseline B*     37,954.9 × 0.951   = 36,095.1
Routine savings            36,095.1 − 19,000  = 17,095.1
Less non-routine adjustment                   −  1,150.0
Verified realised saving                      = 15,945.1   →  $15,950
```

Reconciliation to the Section 3.8 table. Decomposing $B^* - B$ sequentially in equation order (applying the demand factor to B gives $-\$445.03$, and applying the rate factor to the demand-adjusted baseline gives $-\$1{,}859.79$, each rounded to the nearest $\$50$ per Section A.2.3):

	Annualised
Gross observed reduction (B − R)	$19,400
Less: demand normalisation	($450)
Less: provider list-rate reduction	($1,850)
Less: commitment reallocation (non-routine)	($1,150)
Verified realised saving	**$15,950**

A.7.1 What the unit-elasticity assumption would have cost. Had the elasticity been assumed at 1.0 rather than measured at 0.19, the demand normalisation would have deducted **$2,286** rather than **$445**, understating the verified saving by $1,841, or **12% of the reported figure**, from one unexamined modelling assumption. The team would have reported a number that was wrong in the direction of pessimism, and would have had no way to discover it. This term is the strongest practical argument in this appendix for specifying a protocol at all.

Uncertainty. Propagating the standard error on the elasticity through the demand term gives a 95% interval on the verified saving of approximately **$15,600 to $16,300**, or about ±2%. The magnitude of that interval matters less than its size relative to the alternative: the gap between the naive estimate at finding ($24,100) and the verified result ($15,950) is **$8,150**. The protocol's imprecision is roughly 4% of the error it removes. Imprecision in verification is not an argument for reporting the unverified figure.

A.7.2 Option C and the detectability test. Where a change is diffuse (for example, a storage lifecycle policy applied across thousands of buckets), isolation is impossible and the baseline becomes a regression of scope-level cost on drivers. The hazard is that a real but small saving disappears into the scope's own noise, while the regression nonetheless returns a point estimate that looks like a result.

The protocol therefore transposes the compliance threshold of **ASHRAE Guideline 14** (ASHRAE, 2023; Efficiency Valuation Organization, 2022), which requires that **Fractional Savings Uncertainty** (FSU $= \Delta S/S$) not exceed **50% at the 68% confidence level** (1σ). Algebraically, requiring $1 \cdot SE(S)/S \leq 0.50$ is identical to a **two-sigma detectability test**: a saving is reportable under Option C only if its expected magnitude exceeds **twice the standard error of the cumulative reporting-period residual** ($S \geq 2 \cdot SE(S)$), computed from the baseline regression and adjusted for lag-1 serial correlation (ρ_1) in daily cost via ASHRAE Guideline 14's effective sample size $n' = n(1-\rho_1)/(1+\rho_1)$. Illustratively, a scope whose autocorrelation-adjusted baseline regression leaves a residual standard error equivalent to ±$12,000 annualised (so $2 \cdot SE = \$24{,}000$) supports reporting a $96{,}000 expected effect($ = 12.5%$) and does not support reporting an $18{,}000 one($ = 66.7%$). The second is tracked operationally and excluded from financial reporting, not because it is untrue, but because the method cannot distinguish it from the scope's ordinary variation.

A.8 Calibration: closing the loop back to stage 2

Verification's second output is a correction to the estimator. For this finding:

Ratio	Value
Verified divided by naive estimate at finding	0.66
Verified divided by post-evidence estimate	0.84

The second ratio is the useful one, because it measures the estimator the organisation would actually act on. Accumulated across a population of verified findings and segmented by workload class, it yields a class-level realisation factor that stage 2 can apply to future estimates, converting the "verification failed" outcome of Section 3.5 from an embarrassment into the only available training signal.

One observation is not a calibration. A single ratio of 0.84 establishes nothing; the quantity of interest is the distribution of that ratio across findings within a class, and its stability over time. This is the mechanism by which research question 1 in Section 7 becomes answerable from operational data rather than from a survey.

A.9 What this protocol does not solve

1. **The counterfactual remains unobservable.** Every figure the protocol produces is an estimate with a stated method. Verification narrows the error and makes it inspectable; it does not convert an estimate into a measurement of fact.
2. **Elasticity estimates are unstable** where drivers are collinear, where the baseline window is short, or where the workload is itself changing. The protocol offers no minimum window length by workload class, because no evidence supports one.
3. **Shared-platform attribution is unresolved.** The protocol defers to the organisation's existing allocation rules rather than fixing them, and inherits whatever contestation those rules carry.
4. **No inter-rater reliability evidence exists.** Two competent analysts applying this to the same finding may enumerate different non-routine adjustments and report different figures. Pre-registration constrains this but does not eliminate it, and the magnitude of residual disagreement is unknown.
5. **The error bounds in Section A.7 are specific to one synthetic case** and should not be read as characteristic. No distribution of protocol error across real findings or organisations exists.
6. **The Option D exclusion is costly.** It rules prevented and avoided cost, plausibly the largest category of engineering value in a mature practice, out of financial reporting entirely. Some organisations will reasonably judge that trade unacceptable, and the protocol offers them no middle path beyond operational tracking.
7. **It is untested.** No organisation has run this protocol. Its publication here is an invitation to falsify it, and Section 7 states the three questions that would do so.

About the Author

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Continuous Cost Governance is a proposed framework offered for industry review and critique. It is not an industry standard. The framework originated in observations made during the development of a commercial cost governance tool; that origin is disclosed for transparency and the framework is intended to be evaluated on its own merits.